Project Scope

Danish Pharmacy Sales Analysis

Roskilde, Denmark • April 2023

1. Project Background

This project analyses pharmaceutical sales trends for a pharmacy in Roskilde, Denmark, using transaction data from the first 13 days of April 2023. The aim is to identify top-selling medicines, recurring sales patterns, and actionable insights that support inventory management, operational planning, and marketing decisions.

2. Objectives

- Identify top-performing medicines by revenue and units sold.
- Detect temporal patterns including daily, hourly, and weekday-level trends.
- Provide data-driven recommendations for inventory and staffing decisions.
- Forecast future demand using time-series methods to support planning.

3. Dataset Overview

Attribute	Details
Dataset Name	Danish Pharmacy Sales Data — April 2023
Location	Roskilde, Denmark
Period Covered	1 April 2023 – 13 April 2023 (13 days)
Total Records	20,003
Granularity	One record per individual sale transaction

4. Dataset Fields

Field	Description
SaleID	Unique identifier for each sale transaction.

Field	Description
DateTime	Date and time of the sale.
DrugID	Unique identifier for the drug sold.
DrugName	Name of the medicine.
DrugClass	Therapeutic category or drug class.
Price	Unit price of the drug at time of sale.
TotalSales	Revenue generated from the transaction (Price × Quantity).
ExpDate	Expiration date of the drug batch sold.
Quantity	Number of units sold in the transaction.

5. Methodology

The analysis is structured in two phases:

1. **Data Exploration and Cleaning —** Parsing and validating the dataset, handling missing values, and deriving temporal features (date, hour, weekday).
2. **Graphical Analysis —** Visual exploration of revenue and quantity trends, top-medicine rankings, category distributions, and correlation analysis.

6. Future Work

A demand forecasting application has been developed as an extension of this project. The tool employs a weekday-seasonal naive model with exponential smoothing to produce 7- or 14-day demand forecasts at the individual medicine level. Forecast accuracy is evaluated using leave-one-out cross-validation, reporting Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). The application is built with Streamlit and provides interactive medicine selection, forecast visualisation, and CSV export.

7. Limitations

- The dataset spans only 13 days, which restricts the depth of trend analysis and limits the reliability of any predictive model.
- TotalSales represents revenue (Price × Quantity) and cannot be used as a direct proxy for units sold.
- Data originates from a single pharmacy in Roskilde and may not generalise to other locations or pharmacy types.
- External factors such as public holidays, local events, weather, and prescription patterns are not captured in the dataset.